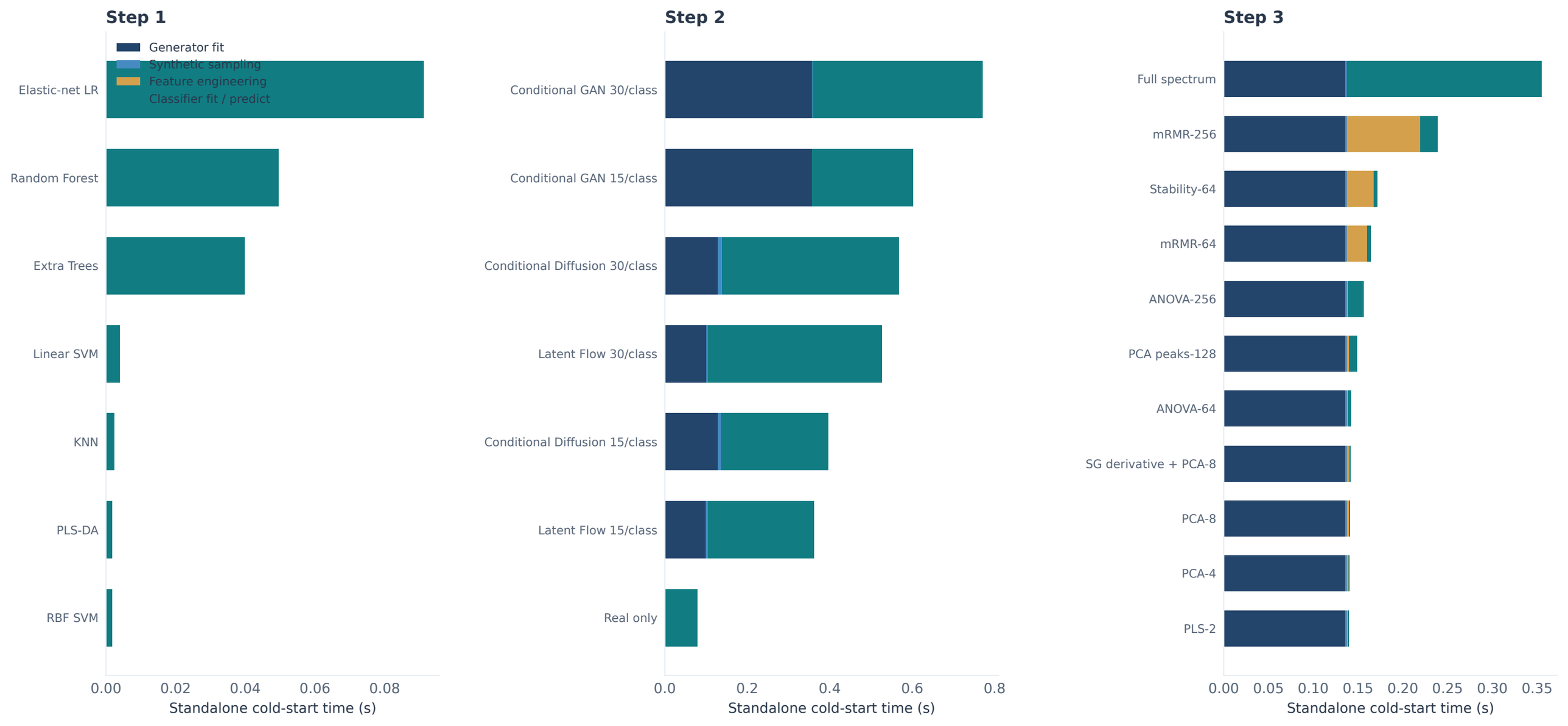
Step 1
Generator fit
Synthetic sampling
Feature engineering
Classifier fit / predict
Elastic-net LR
Random Forest
Extra Trees
Linear SVM
KNN
PLS-DA
RBF SVM
0.00
0.02
0.04
0.06
0.08
Standalone cold-start time (s)
Step 2
Conditional GAN 30/class
Conditional GAN 15/class
Conditional Diffusion 30/class
Latent Flow 30/class
Conditional Diffusion 15/class
Latent Flow 15/class
Real only
0.0
0.2
0.4
0.6
0.8
Standalone cold-start time (s)
Step 3
Full spectrum
mRMR-256
Stability-64
mRMR-64
ANOVA-256
PCA peaks-128
ANOVA-64
SG derivative + PCA-8
PCA-8
PCA-4
PLS-2
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Standalone cold-start time (s)